2026 | Version 1.0

Global Supply Chain & Logistics Visibility Platform

Enterprise Architecture & Systems Analysis

C4 Model | EDA | BPMN | UML | API Specs

PROJECT HIGHLIGHTS

15 Enterprise Diagrams | Event-Driven Architecture | Microservices

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Document Control

Item	Detail
Document Title	Global Supply Chain Visibility Platform — Enterprise Architecture
Version	1.0
Author	Abdullah Ibrahim Elsayed Mohamed Tawfik — Systems Analyst
Audience	Hiring Managers, Solution Architects, Enterprise Architects, Logistics IT Recruiters
Purpose	Portfolio Case Study demonstrating enterprise architecture and event-driven systems design
Confidentiality	Public (portfolio) — no real customer or carrier data used
Status	Final — Approved for portfolio distribution

Revision History

Version	Date	Author	Change
0.1	Aug 2026	Abdullah Ibrahim Elsayed Mohamed Tawfik	Initial C4 Context + Event-Driven Architecture design
0.5	Sep 2026	Abdullah Ibrahim Elsayed Mohamed Tawfik	Added 15 enterprise diagrams (C4, BPMN, UML, ERD)
1.0	Oct 2026	Abdullah Ibrahim Elsayed Mohamed Tawfik	Final — full architecture document with API specifications and NFRs

This document follows a standard revision workflow (Draft → Internal Review → Baseline) consistent with enterprise documentation practices for large-scale distributed systems.

1. Executive Summary

Modern supply chains are highly fragmented. Shippers, carriers, customs authorities, and warehouses operate in silos using disparate systems, spreadsheets, and emails. This lack of visibility leads to unpredictable ETAs, lost shipments, and delayed customs clearance.

This document presents a complete enterprise architecture and systems analysis for a Global Supply Chain Visibility Platform (SCVP). The platform acts as a centralized 'Control Tower' that ingests millions of events per second from IoT sensors and carrier APIs, processes them using an Event-Driven Architecture (EDA), and provides real-time tracking, predictive ETAs, and automated exception management.

1.1 The Problem

- Lack of real-time visibility into shipments across different carriers.
- Reactive exception handling (e.g., customs holds, temperature drops) instead of proactive.
- High latency in data integration (batch processing vs. real-time webhooks).
- No single source of truth for supply chain KPIs.

1.2 Business Objectives

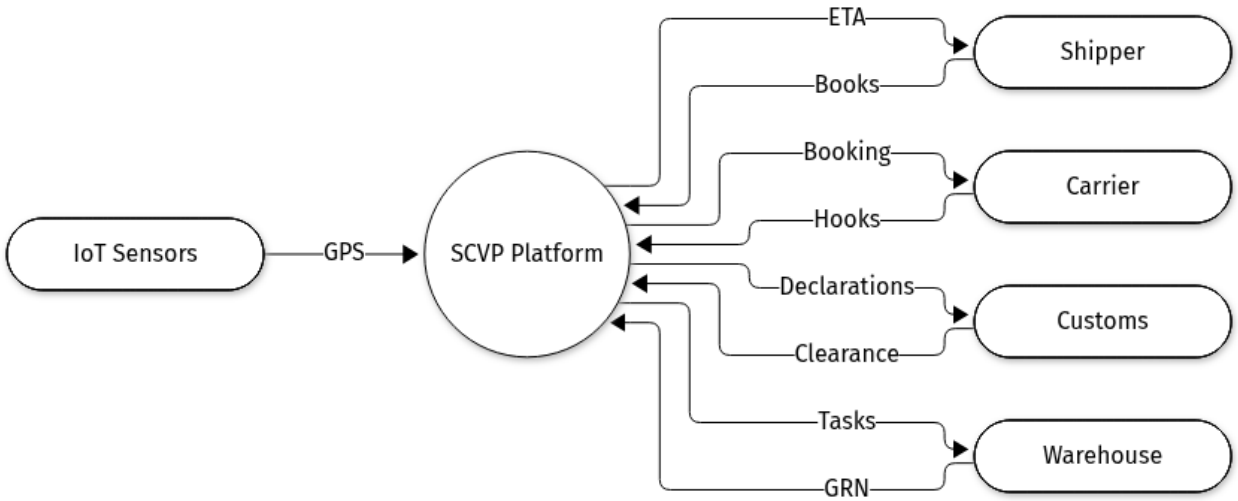
ID	Objective	Current	Target
BO-01	Real-time tracking latency	Batch (Hours)	Under 500ms
BO-02	Exception resolution time	Manual / Days	Automated / Minutes
BO-03	ETA Accuracy (Predictive)	65%	95%
BO-04	API Integration Uptime	95%	99.99%
BO-05	Operational Cost	Baseline	Reduce 30%

2. System Architecture (C4 Model)

The platform is designed using the C4 model for visualizing software architecture, ensuring scalability and separation of concerns.

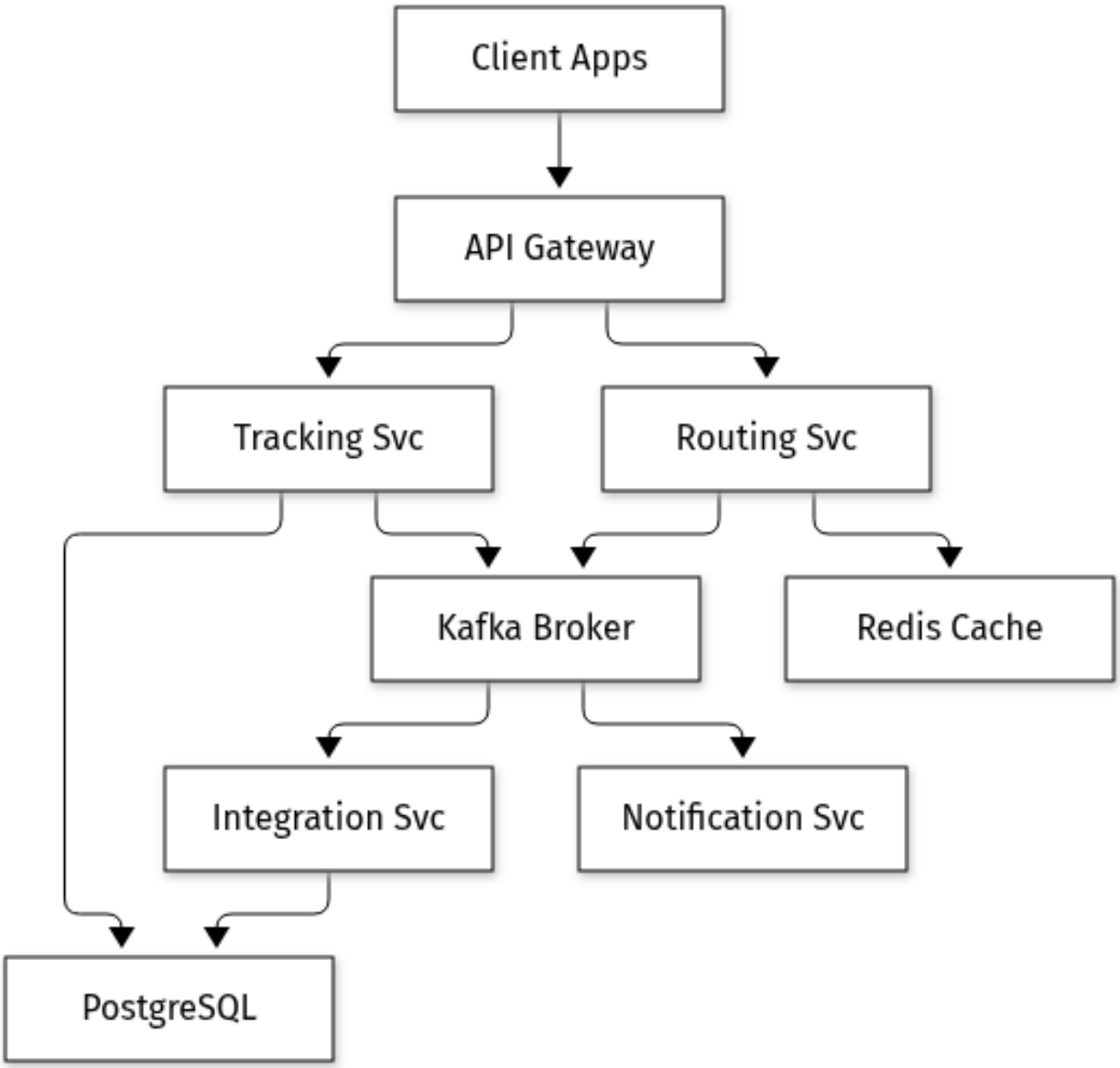
2.1 Context Diagram (C4 Level 1)

Shows the SCVP platform interacting with external actors (Shippers, Carriers, Customs, IoT).



2.2 Container Diagram (C4 Level 2)

Zooms into the platform, revealing Microservices, Kafka, and Databases.



2.3 Event-Driven Architecture (EDA) Flow

How IoT data and Carrier Webhooks flow through Kafka to the Dashboard.



3. Requirements Specification

3.1 Functional Requirements (FRs)

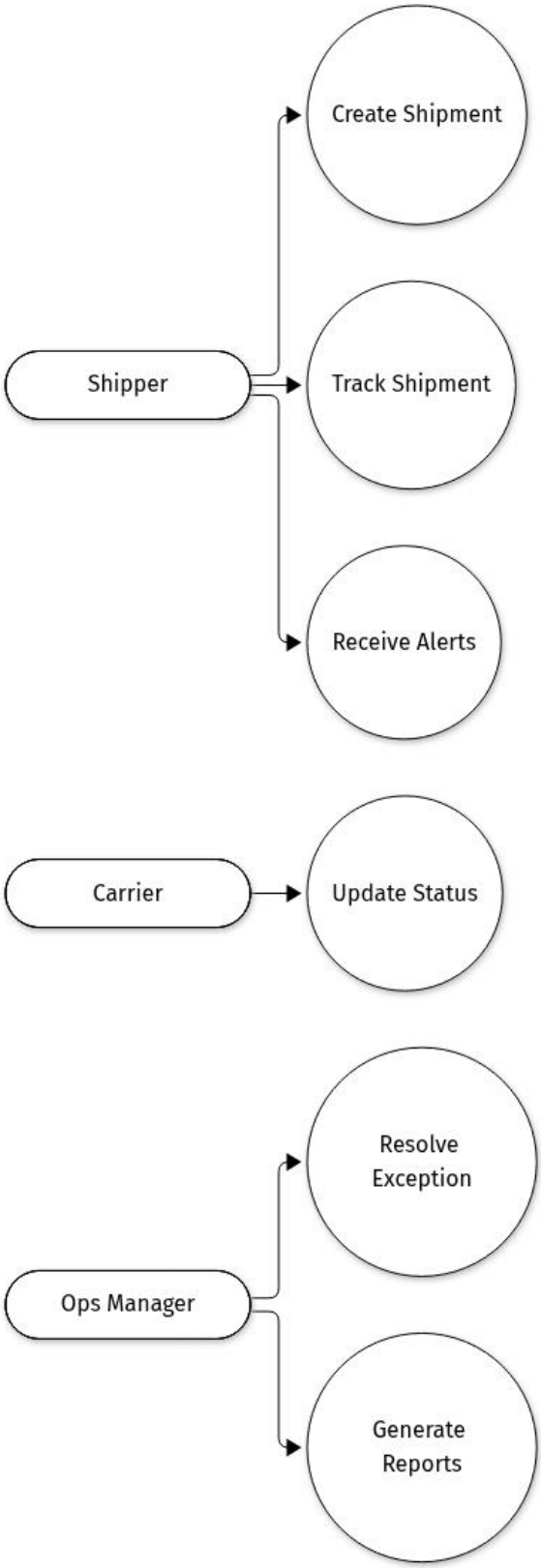
ID	Requirement	Priority
FR-01	Ingest high-throughput IoT telemetry (GPS, Temperature).	Must
FR-02	Receive and process Carrier Webhooks (idempotent).	Must
FR-03	Calculate Predictive ETA using machine learning models.	Must
FR-04	Trigger automated alerts on threshold breaches (e.g., Temp > 8°C).	Must
FR-05	Provide a unified dashboard for tracking all shipments.	Must
FR-06	Integrate with Customs APIs for pre-clearance filing.	Must
FR-07	Generate automated Exception Tickets for Ops team.	Should
FR-08	Expose REST API for Shippers to query shipment status.	Must

3.2 Non-Functional Requirements (NFRs)

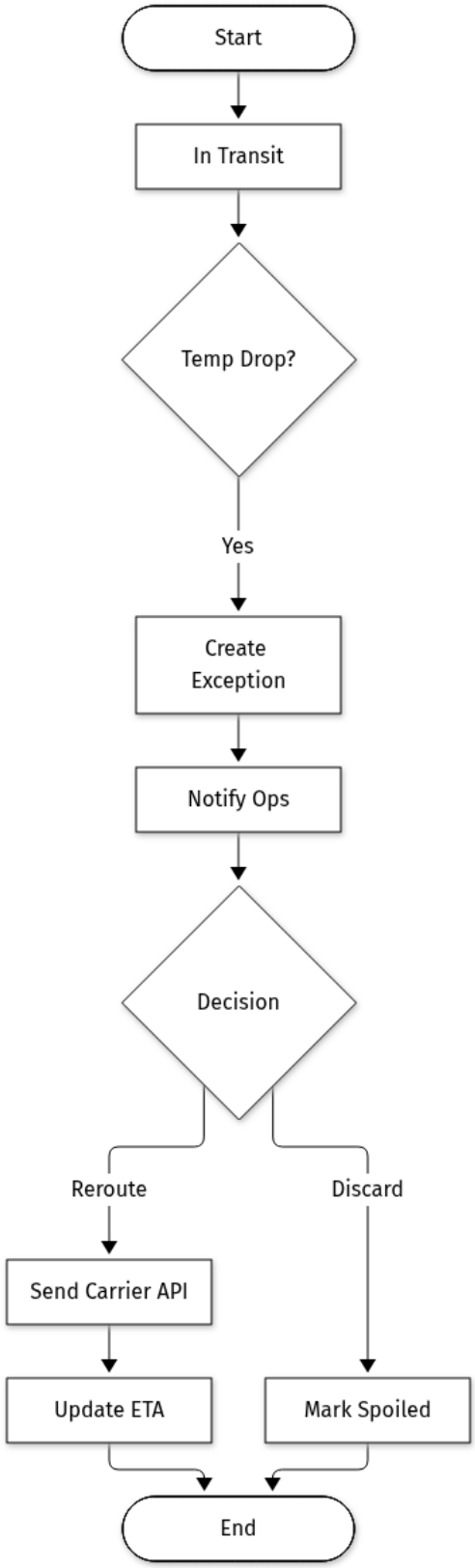
ID	Category	Requirement
NFR-01	Scalability	Handle 10,000 events per second (EPS) peak load.
NFR-02	Latency	End-to-end event processing < 500 milliseconds.
NFR-03	Availability	99.99% uptime for the Tracking API.
NFR-04	Resilience	Circuit Breaker pattern for all external integrations.
NFR-05	Consistency	Eventual consistency for Tracking data via Kafka.
NFR-06	Security	OAuth2, mTLS for IoT devices, PII encryption at rest.

4. Analysis Diagrams

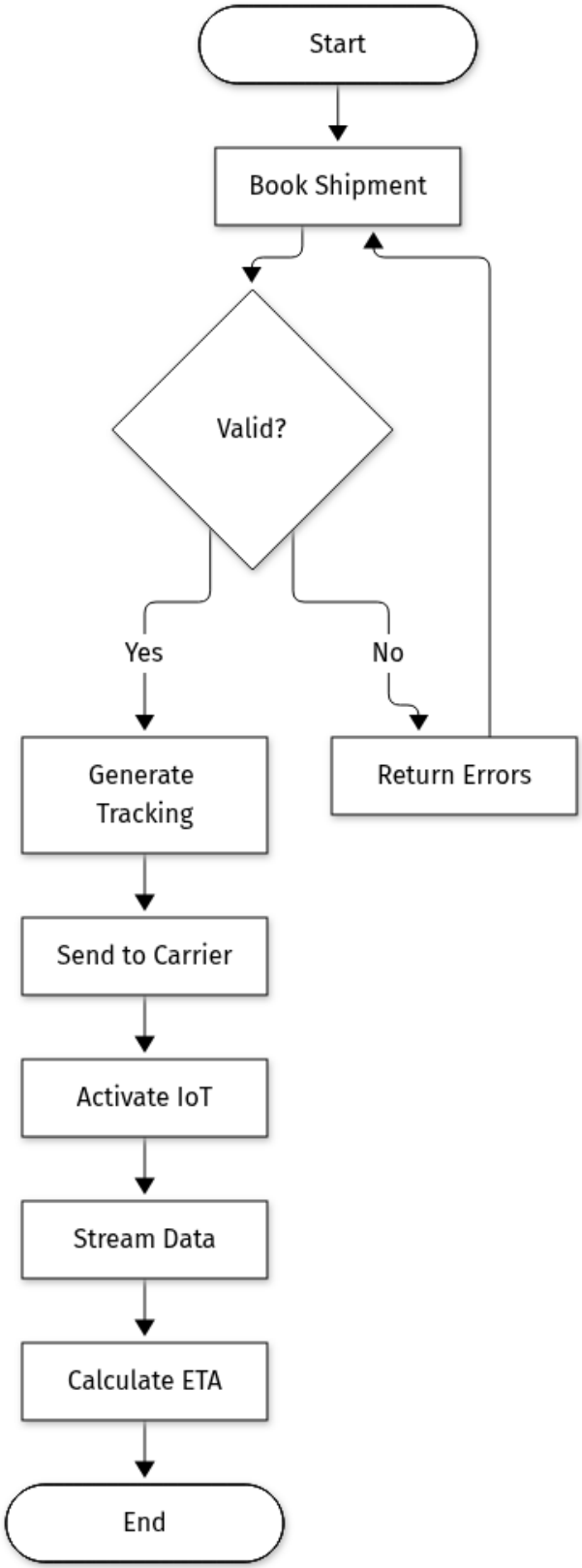
4.1 Use Case Diagram



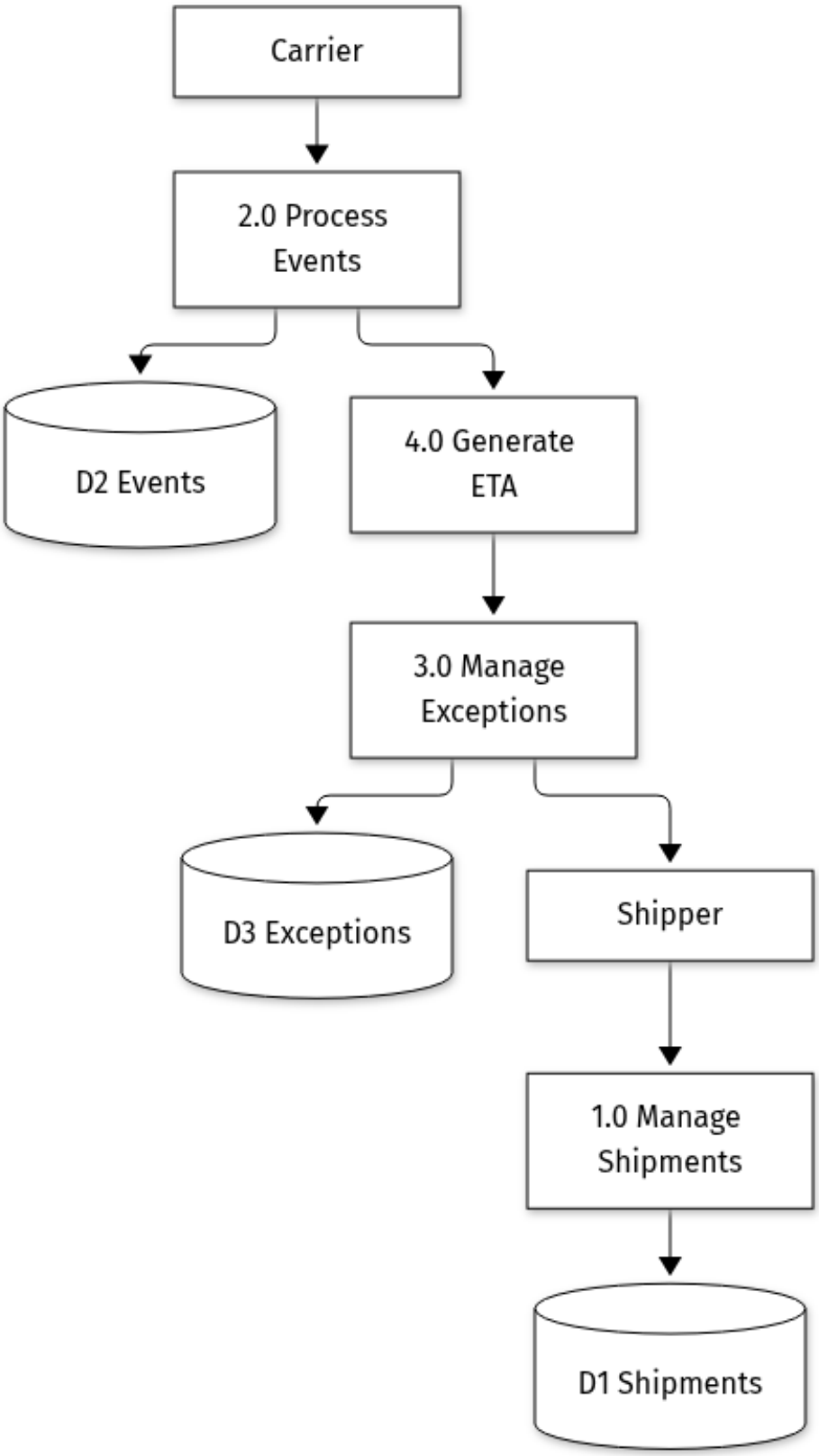
4.2 BPMN: Exception Management Process



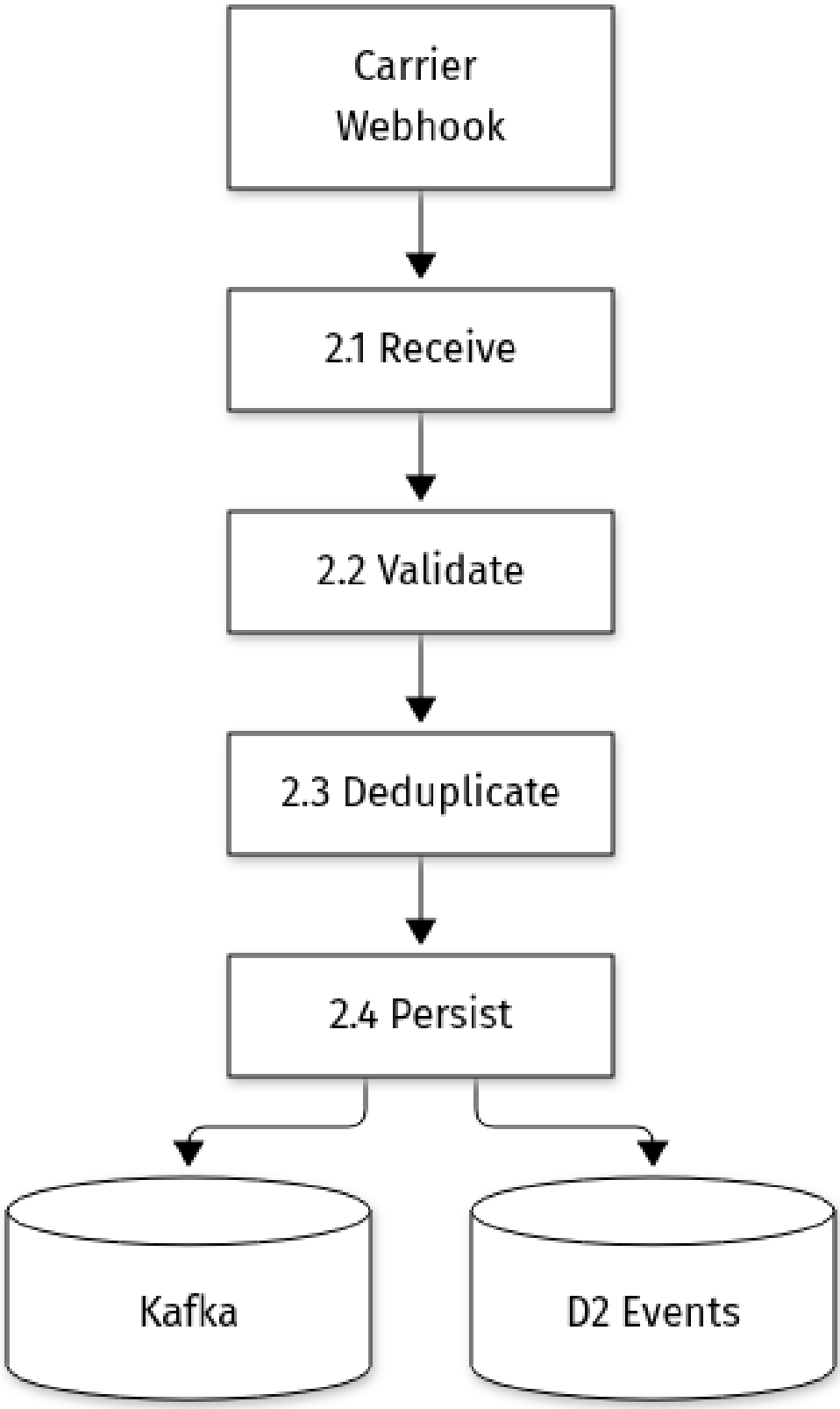
4.3 Activity Diagram: Shipment Booking



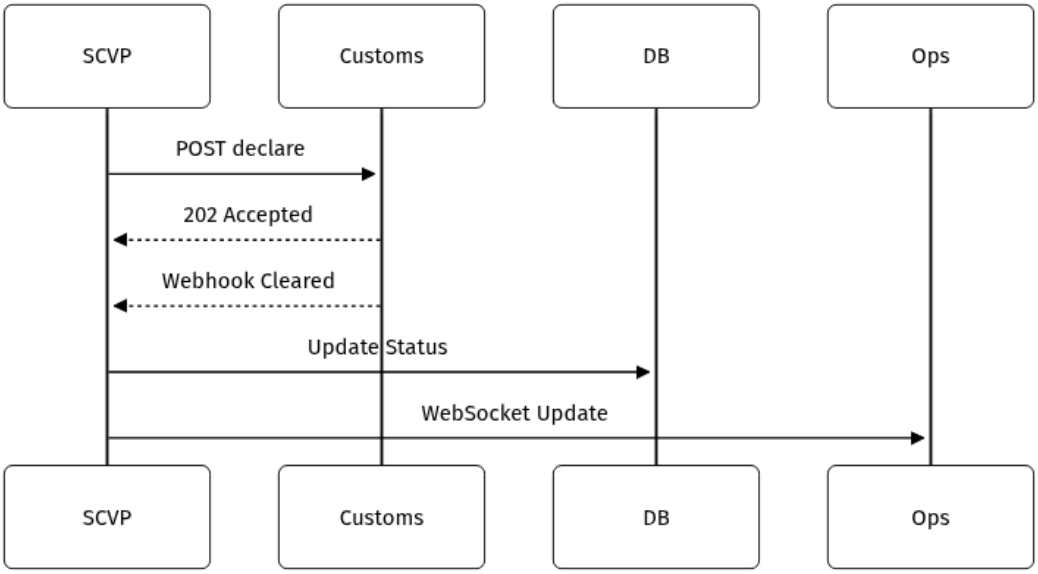
4.4 Data Flow Diagram (Level 1)



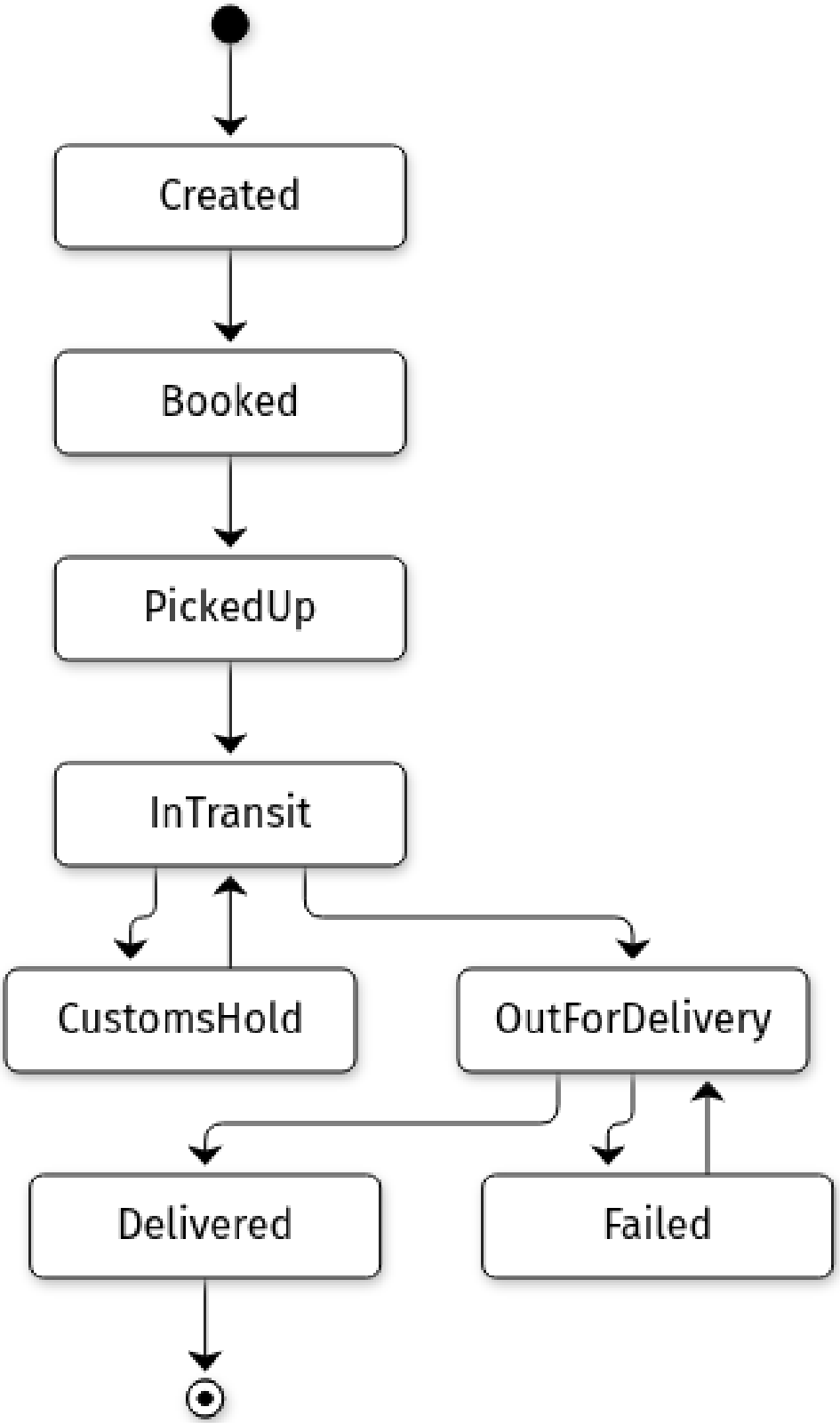
4.5 Data Flow Diagram (Level 2 - Event Processing)



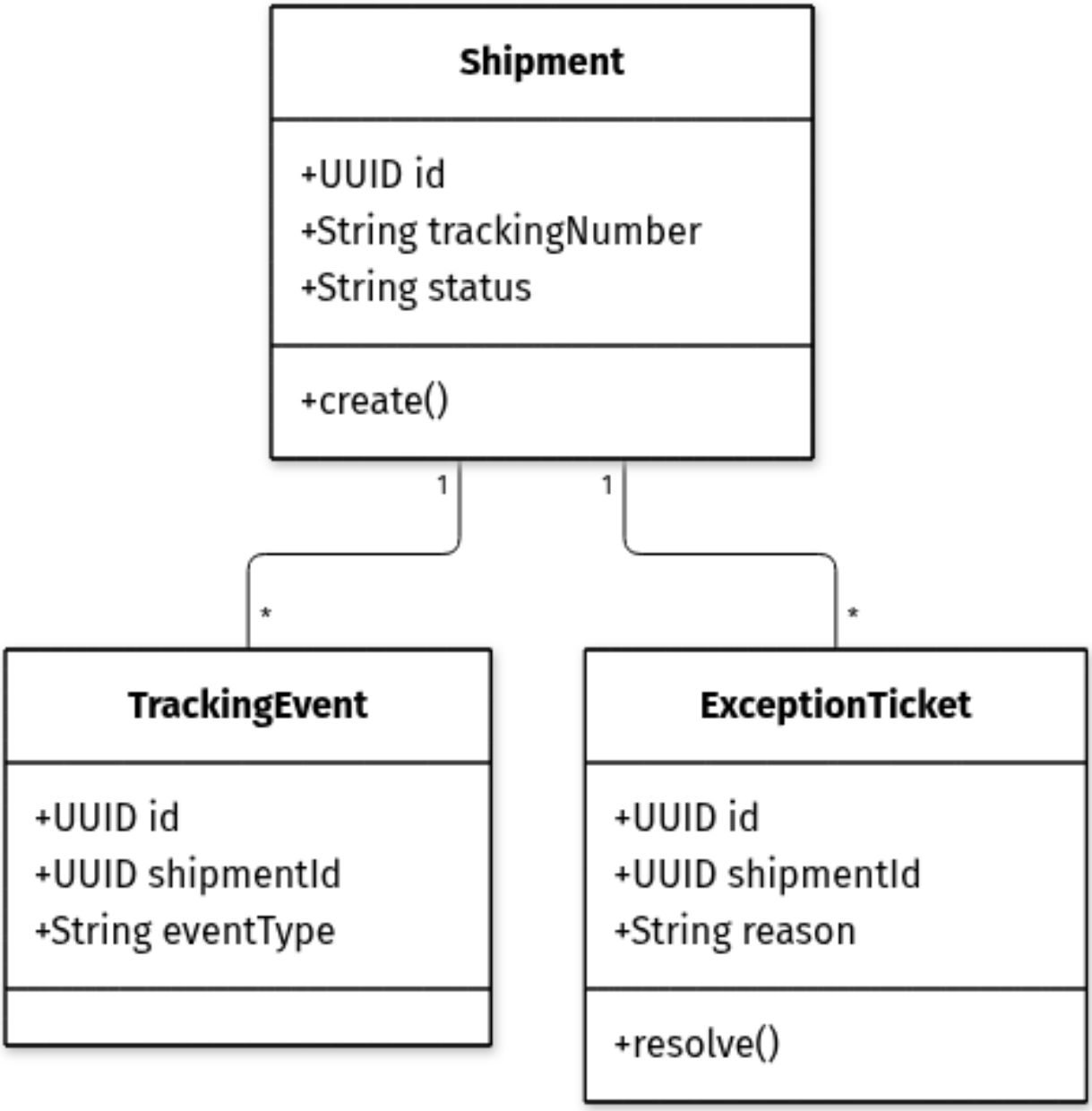
4.6 Sequence Diagram: Customs Integration



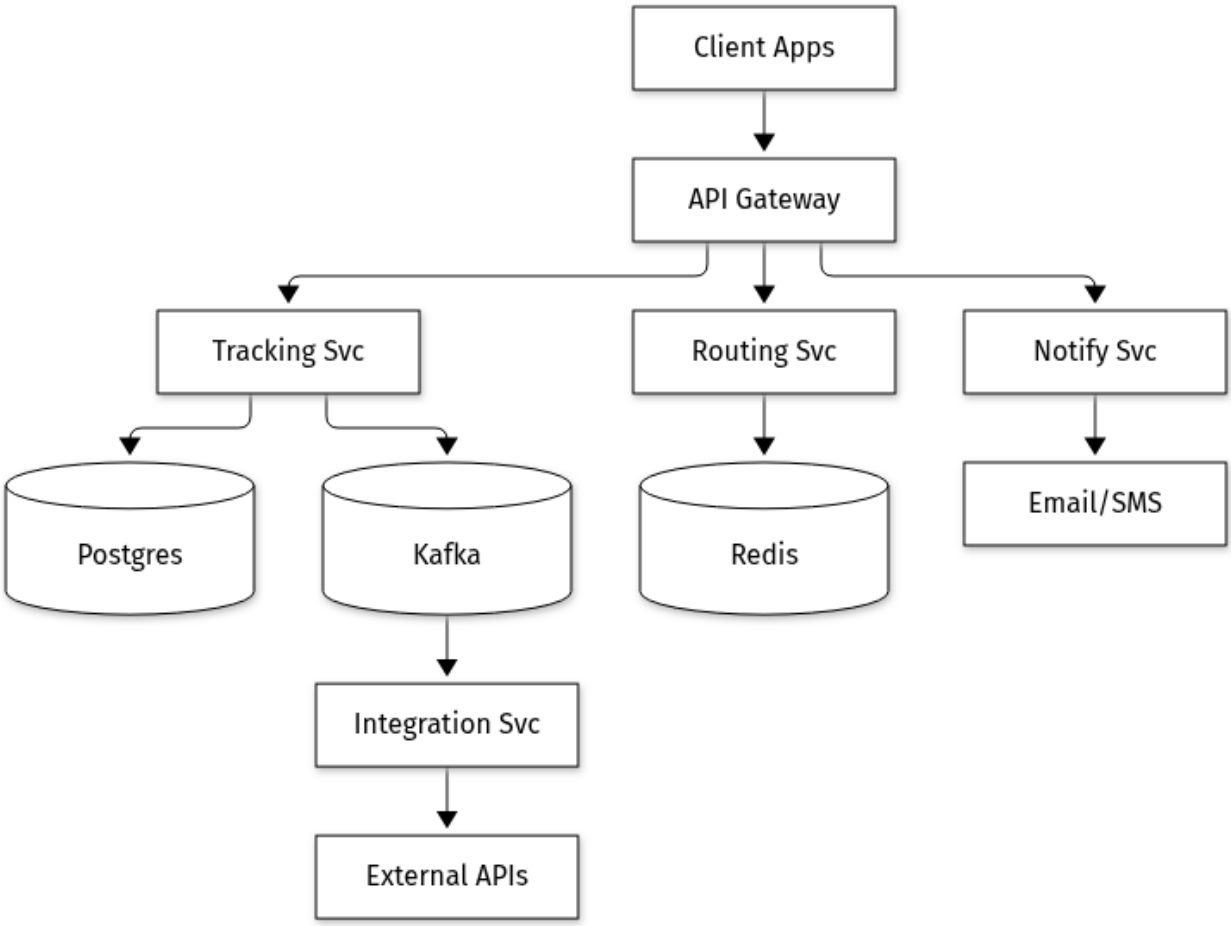
4.7 State Diagram: Shipment Lifecycle



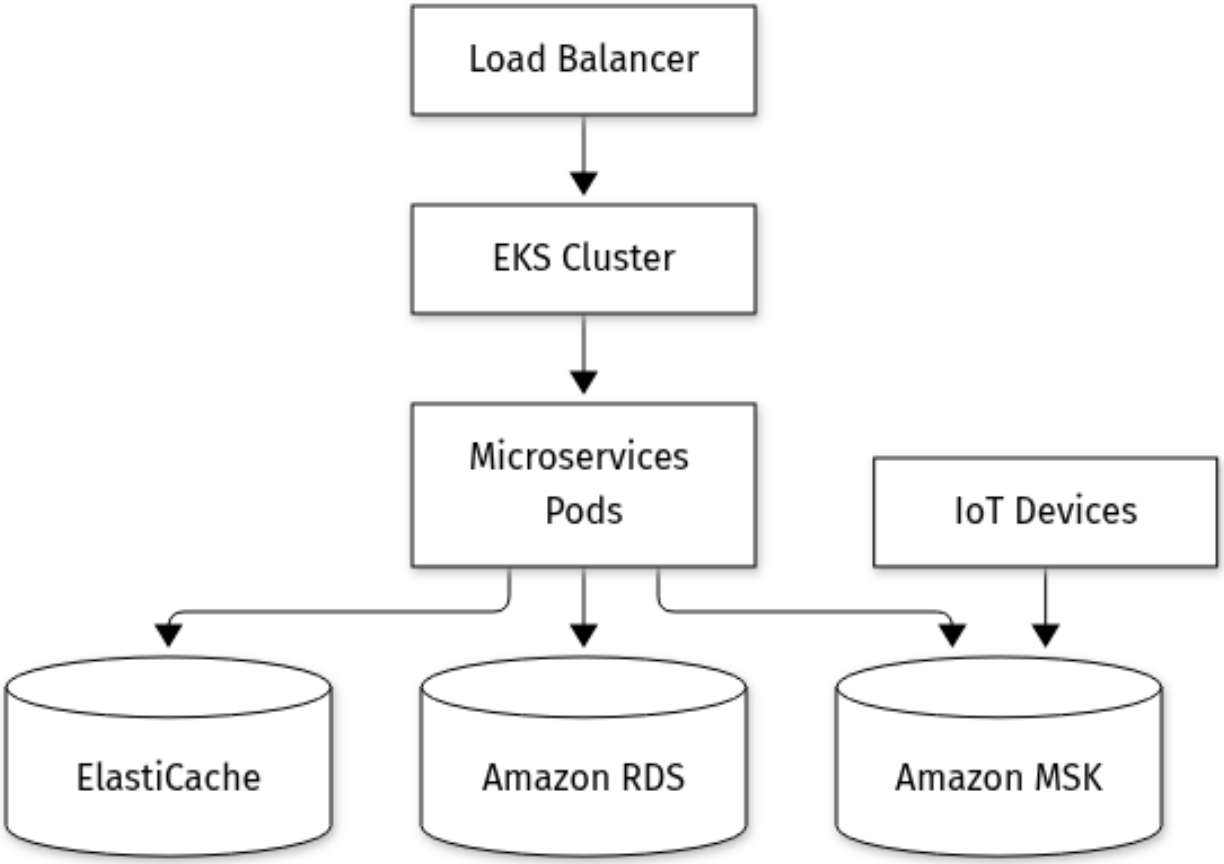
4.8 Class Diagram (Domain Model)



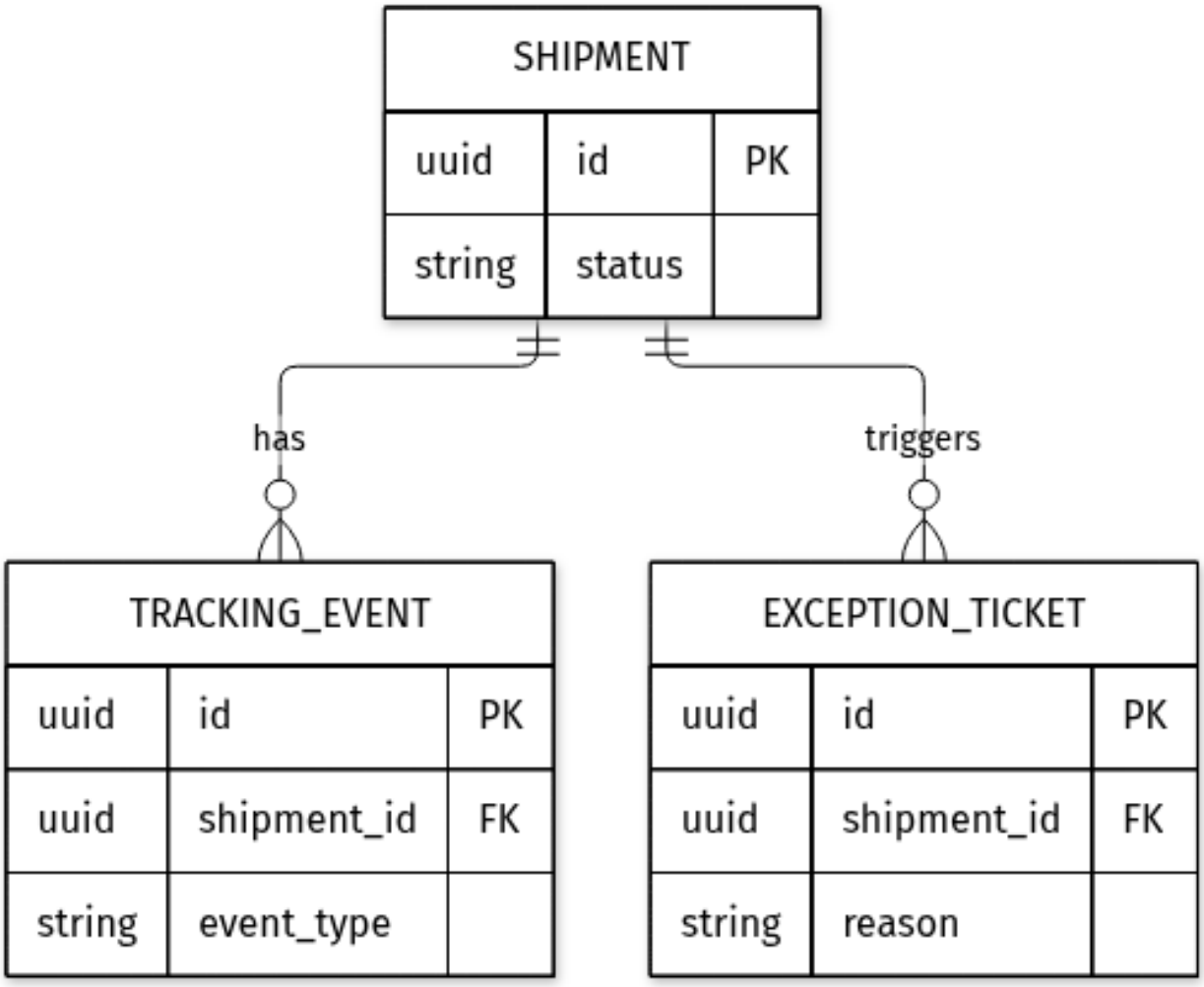
4.9 Component Diagram (Microservices)



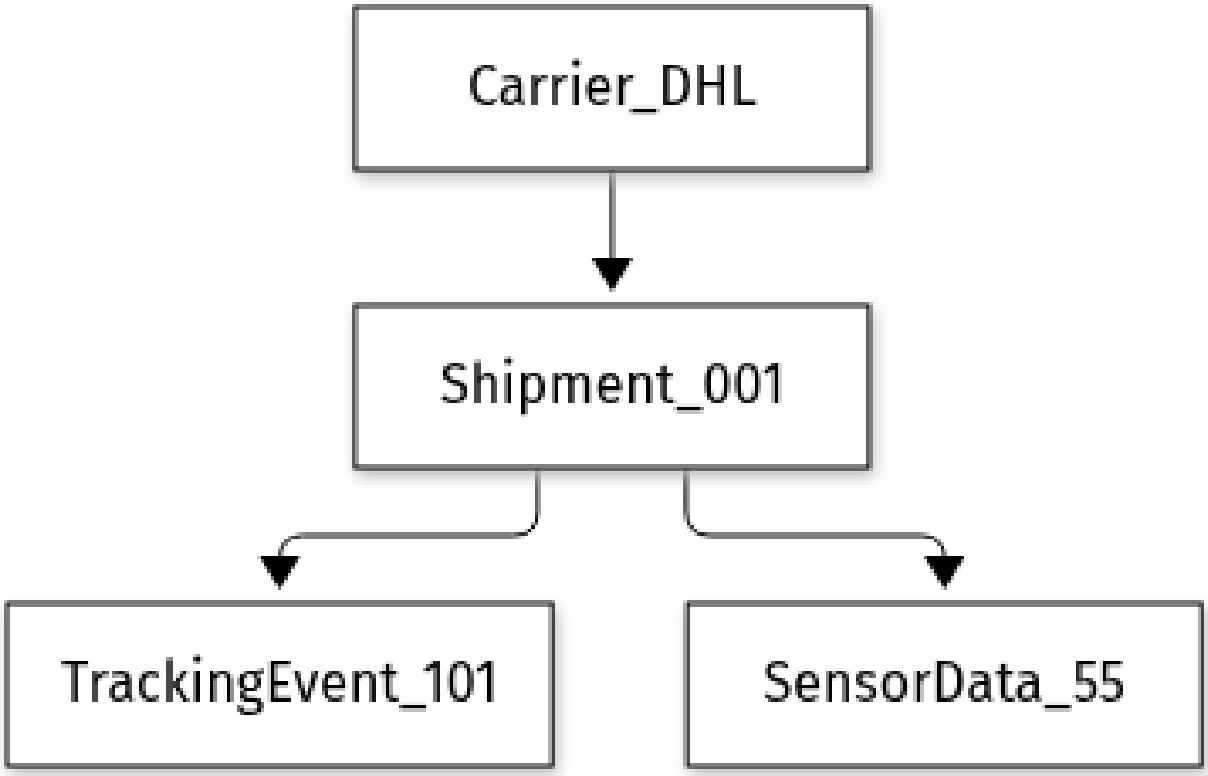
4.10 Deployment Diagram (AWS Cloud)



4.11 Entity Relationship Diagram (ERD)



4.12 Object Diagram (Runtime Instance)



5. API & Integration Specifications

The platform exposes REST APIs for synchronous queries and uses Webhooks for asynchronous updates. All external integrations must implement Idempotency and Circuit Breaking.

5.1 Carrier Webhook Payload (Inbound)

```
{
  "event_id": "evt_9982a1b",
  "tracking_number": "TRK-88211",
  "event_type": "DEPARTED_ORIGIN",
  "timestamp": "2026-11-01T10:00:00Z",
  "location": {
    "lat": 30.0444,
    "lng": 31.2357
  },
  "carrier_code": "DHL"
}
```

5.2 Shipper Tracking API (Outbound)

```
GET /v1/shipments/{tracking_number}/status
Headers: Authorization: Bearer <token>
```

```
{
  "tracking_number": "TRK-88211",
  "status": "InTransit",
  "eta": "2026-11-03T14:00:00Z",
  "last_event": "DEPARTED_ORIGIN",
  "sensor_health": "OK"
}
```

6. Analytical Tables

6.1 Risk Analysis Matrix

ID	Risk	Probability	Impact	Mitigation
R-01	Kafka Cluster Failure	Low	Critical	Multi-AZ deployment, automated failover.
R-02	Carrier API Rate Limiting	High	Medium	Implement exponential backoff and Redis caching.
R-03	IoT Data Spoofing	Low	High	mTLS certificates for all edge devices.
R-04	Duplicate Webhooks	High	Low	Enforce idempotency keys at API Gateway.
R-05	Predictive ETA Drift	Medium	Medium	Continuous model retraining with historical data.

6.2 Traceability Matrix (Sample)

Requirement	Architecture Component	Diagram Ref	API Ref
FR-01 (IoT)	Kafka / Tracking Service	2.2, 2.3	5.1
FR-02 (Webhooks)	Integration Service	2.2, 4.6	5.1
FR-03 (ETA)	Routing Service	4.8	5.2
FR-07 (Exceptions)	Exception Ticket DB	4.2, 4.7	-

7. Glossary

Term	Meaning
SCVP	Supply Chain Visibility Platform
EDA	Event-Driven Architecture
C4 Model	Context, Containers, Components, Code (Architecture framework)
Kafka	Distributed Event Streaming Platform
ETA	Estimated Time of Arrival
mTLS	Mutual Transport Layer Security
Idempotency	Ensuring an operation can be applied multiple times without changing the result

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